

Literature review for automated ECG change detection during concurrent EEG monitoring in epilepsy

Executive synthesis

This project sits in a real and important gap. An International League Against Epilepsy ¹ guideline defines long-term video-EEG monitoring as scalp EEG plus a single ECG channel, so the acquisition setup you want to study is already part of routine epilepsy care. At the same time, a 2024 review on concurrent ECG-EEG argued that routine dual monitoring in epilepsy units needs more standardized protocols and explicitly pointed to future wearable and computer-based monitoring as a major next step. A separate systematic review focused on ECG findings beyond heart rate concluded that most epilepsy ECG work has historically concentrated on heart rate and heart-rate-related variables, whereas conduction and morphology changes remain much less developed for automation. In other words, the clinical importance is established, the signal source is already present in EMU workflows, and the automation problem is still open. ²

The literature is strongest for rate and variability, reasonably mature for rhythm surveillance, and still comparatively thin and methodologically cautious for single-lead morphology. That pattern matters for scope. A meeting-ready summary would be: **automated peri-ictal HR and HRV detection is clearly supported by prior work; automated arrhythmia transition detection is plausible by borrowing from ambulatory ECG practice; automated ST/T/QT/QRS analysis is feasible as within-patient trending, but should be framed more cautiously than a 12-lead diagnostic interpretation problem.** That is probably the most defensible way to describe the project's novelty and its achievable first milestones. ³

Clinical substrate for the detector

The core physiologic target is not hypothetical. In a major review of seizure-related cardiac manifestations, ictal tachycardia was described as the most common peri-ictal cardiac finding, reported in about 82% of patients on average. Across studies, heart rate changes during seizures were reported to average roughly 30 bpm or more than 50% above interictal heart rate, usually occurring in the pre-ictal period or within 30 seconds after seizure onset, with peak heart rate reached within the first minute for most seizures. The same review notes that heart-rate change appears in both generalized tonic-clonic and focal seizures, and is especially frequent in temporal lobe epilepsy. Foundational video-EEG work by Zijlmans and colleagues also showed that heart-rate increase of at least 10 bpm occurred in 73% of seizures, that in 23% of seizures the increase preceded both electrographic and clinical onset, and that ECG abnormalities were present in 26% of seizures and 44% of patients. ⁴

The rhythm and morphology targets in your project are also directly grounded in the literature. The same epilepsy review reports that seizures can be accompanied by atrioventricular block, atrial fibrillation, supraventricular tachycardia, atrial and ventricular premature depolarizations, ventricular fibrillation, and QT shortening or prolongation. It also states that direct ischemia-like myocardial effects can occur, with ST-

segment depression reported in up to 40% of seizures, while seizure-related Takotsubo syndrome is a recognized post-seizure phenomenon, especially after generalized tonic-clonic seizures or status epilepticus. A dedicated review of arrhythmias during and after seizures further emphasizes that postictal arrhythmias are seen especially after convulsive seizures and are often the cases most closely linked to near-SUDEP and SUDEP mechanisms. ⁵

That clinical substrate is one reason the project is relevant to SUDEP, but the literature also shows why the detector must not reduce everything to heart rate alone. The 2021 review links severe peri-ictal bradyarrhythmias and asystole to postictal hypoxemia and respiratory dysfunction, and notes that in the MORTEMUS work abnormal heart rhythms appeared after apnea onset in all deceased patients studied during long-term video-EEG. The implication is important for your framing: the detector should ideally characterize a **sequence** of peri-ictal changes across rate, rhythm, morphology, and variability, not just a single tachycardia threshold. ⁶

What the concurrent ECG-EEG and EMU literature already shows

The dedicated EMU literature is not empty, but it is fragmented. The classic concurrent video-EEG/ECG studies established the phenomenon first and the automation problem only indirectly. Zijlmans et al. analyzed prolonged video-EEG with two-channel ECG and showed that heart-rate changes often cluster tightly around, or even before, seizure onset. Chen et al. later analyzed 181 partial seizures from 81 presurgical patients and found ictal tachycardia in 83.98% of seizures, with only one ictal bradycardia case. They also showed that HR increase was independently associated with seizure side, seizure type, and sleep-wake state, with more marked increases in right-sided foci, male patients, sleep, and secondarily generalized complex partial seizures. HR rose rapidly within the 10 seconds before seizure onset and during ictus, then usually slowed toward baseline with seizure offset. For your project, that is strong justification for EEG-timed windows and for explicit state conditioning. ⁷

The pre-ictal literature reinforces this. A systematic review and meta-analysis by Bruno et al. included 30 studies, 1,110 participants, and 2,957 seizures and estimated a pooled incidence of pre-ictal heart-rate increase of 36 per 100 seizures, rising to 44 per 100 seizures in temporal lobe epilepsy and 55 per 100 in adult cohorts. That does not mean every seizure produces a stable pre-ictal signature, but it does mean your proposed segmentation into interictal, pre-ictal, ictal, and post-ictal windows is directly supported by prior evidence rather than being an arbitrary machine-learning convenience. ⁸

What appears much less standardized is the use of concurrent ECG-EEG for **automated** waveform-based interpretation beyond rate. The 2024 review on the “brain-heart connection” argues that dual ECG/EEG recording is valuable both ictally and interictally, and calls for standardized concurrent monitoring approaches in EMUs. The 2020 systematic review of ECG features beyond heart rate screened 502 abstracts, read 110 full papers, and included only 24 papers in the final review, which is a useful quantitative sign that the beyond-HR literature is still relatively small. A fair synthesis for your meeting is therefore: **the neurologic literature clearly proves the phenotype, but has not yet produced a dominant, standardized algorithmic pipeline for single-lead EMU ECG morphology and rhythm analytics.** ⁹

Preprocessing and quality control architecture from the literature

The preprocessing stage you described is exactly where the literature says the problem becomes methodologically serious. Baseline wander is a low-frequency ECG artifact produced by breathing, electrode effects, and subject movement, and it matters especially when subtle morphology endpoints such as ST deviation are being measured. The ST-preservation literature emphasizes that baseline-wander removal is necessary, but that overly aggressive filtering can attenuate or distort genuine ST changes. In parallel, recent single-lead QRS benchmarking work stresses that QRS detection is a foundational step for downstream arrhythmia analysis and becomes substantially harder in noisier telehealth or ambulatory recordings. The practical consequence is that your preprocessing cannot be “filter, then detect peaks” in a generic sense; it has to be designed around preserving morphology while extracting robust beat timing in noisy windows. ¹⁰

One of the most useful adjacent literatures here is automatic ECG signal-quality assessment. A 2022 systematic review of automatic ECG quality assessment techniques concluded that noise remains a major reason automated ECG analysis fails and that the field still needs clearer methods before full clinical translation. Even more practical is a 2022 single-lead wearable ECG quality study that used three quality levels rather than a binary good/bad label: high-quality windows allowed reliable P, QRS, ST, and T assessment; medium-quality windows preserved reliable QRS detection and heart-rate extraction but left other landmarks unclear; low-quality windows needed rejection. That triage logic maps almost perfectly onto your project. It suggests that ECG quality masks should be **domain-specific**: some windows are good enough for rate, fewer for rhythm, and fewer still for morphology. ¹¹

This point becomes even more important around convulsive seizures. In a 2024 multicenter study of 386 seizures from 111 individuals across three EMUs, motor seizure elements significantly impaired some wearable biosignals and accelerometry correlated negatively with peri-ictal signal quality. Although that paper studied wrist biosignals rather than EMU ECG alone, its take-home message transfers: seizure movement itself degrades physiologic signal quality, which means convulsive epochs need explicit artifact logic rather than blind morphology calling. For your detector, that supports a design in which motion-heavy or poor-contact windows can still contribute **rate** features if QRS detection survives, while **morphology** features are suppressed unless signal quality clears a higher threshold. ¹²

Designing detection across rate, rhythm, morphology, and variability

The **rate** domain is where the literature is deepest and the design path is clearest. The seizure literature supports beat-to-beat RR analysis, instantaneous HR, relative HR increase, and derivatives of rate acceleration around seizure onset. The adjacent vagus nerve stimulation literature is especially informative because it operationalizes seizure-linked tachycardia in a device context. Official device documentation and FDA labeling describe AutoStim as detection of rapid **relative** heart-rate increases over baseline, typically programmable from 20% to 70%, with settings chosen at or below the patient’s typical seizure-associated increase; the same materials explicitly warn that exercise and normal autonomic HR changes during wake or sleep can create false positives. The E-37 trial literature similarly reported that at least one AutoStim threshold could exceed 80% sensitivity for seizures accompanied by at least 20% ictal HR increase. Conceptually, that is very close to what your rate detector should do: use patient-specific, state-aware

relative change rather than one universal tachycardia cutoff. Official materials from LivaNova ¹³ and the U.S. Food and Drug Administration ¹⁴ make that principle unusually concrete. ¹⁵

The **rhythm** domain should borrow heavily from ambulatory ECG practice. The ambulatory ECG consensus literature describes modern devices as beat-to-beat systems with automatic arrhythmia detection and, across device classes, available analyses that can include arrhythmia analysis, ST analysis, HRV, and QT dynamics. Patch monitors use beat-by-beat QRS detection as the base layer for rhythm interpretation, whereas event and loop recorders add autotrigger logic for intermittent abnormalities. That suggests a natural epilepsy adaptation: use robust beat detection first, then classify transitions away from a patient's baseline sinus rhythm into bradycardia, pause/asystole, ectopy, supraventricular tachyarrhythmia, or other arrhythmia candidates, with seizure timing used later as context rather than as a substitute for cardiac event logic. The same ambulatory literature is also a reminder that arrhythmia surveillance has already solved many engineering questions that epilepsy research does not need to reinvent. ¹⁶

The **morphology** domain is clinically important but should be framed with the most methodological caution. Major ECG standardization and ambulatory ECG statements from the American Heart Association ¹⁷, the Heart Rhythm Society ¹⁸, and the International Society for Holter and Noninvasive Electrocardiology ¹⁹ emphasize that interval measurements are best obtained from time-coherent multiple leads, that QT prolongation flagged by automatic algorithms should be visually validated, and that single-lead recordings lack the spatial orientation needed for localization of P, QRS, ST, and T wave changes. At the same time, a 2022 patch-sensor validation study reported that small bipolar single-lead ECG can preserve reasonable morphology and timing accuracy relative to standard ECG. The most defensible interpretation is that single-lead EMU morphology analysis is best pursued as **within-patient trending** of ST level, T-wave amplitude, QT/QTc, and QRS duration, preferably with persistence criteria and adjudication subsets, rather than as a stand-alone ischemia diagnosis engine. That is an inference from the standards and validation literature, but it is a strong one. ²⁰

The **variability** domain is where your control cohort and sleep-state conditioning become essential. The classic HRV standards define time-domain, frequency-domain, and nonlinear approaches, while epilepsy-specific guidance argues that inconsistent HRV protocols have slowed clinical translation and recommends a minimum protocol using 5-minute rMSSD measurements in wakefulness and light sleep. Sleep physiology reviews show that the wake-to-NREM transition is associated with increased parasympathetic modulation and reduced sympathetic tone, whereas NREM-to-REM shifts toward vagal withdrawal and relative sympathetic predominance. Separate sleep-engineering literature has shown that HRV alone can classify sleep stages at moderate but meaningful accuracy in large polysomnography datasets. The direct implication for your detector is that normal wake/sleep physiology is not nuisance variance to be ignored; it is a structured confound that should either be modeled explicitly from EEG sleep staging or approximated via ECG-derived sleep-state features. ²¹

The adjacent literatures most worth borrowing from

The first adjacent literature is **polysomnography and sleep staging**. Your description that “similar analyses have been done during polysomnography” is correct. HRV-based and ECG-based sleep-state classification has been studied for years, and modern deep temporal models trained on hundreds of nights can classify sleep stages from HRV with useful accuracy. That matters because it offers a ready-made strategy for handling one of the biggest confounders in peri-ictal ECG work: normal physiological HRV and rate shifts across wake, NREM, and REM. Even if your final detector uses EEG sleep labels rather than ECG-derived

sleep labels, the PSG literature gives you a methodological precedent for state-conditioned baselines and windowed comparison logic. ²²

The second adjacent literature is **Holter, event monitoring, and loop recorders**. Ambulatory ECG devices have already converged on a modular design philosophy: continuous recording when events are frequent, event or loop recording when events are infrequent, autotrigger logic for transient arrhythmias, and domain-specific analyses including rhythm, ST, HRV, and QT trending. The epilepsy review literature even notes that implantable loop recorders may sometimes be required to distinguish ictal syncope or ictal asystole from a primary cardiac event. For your project, that is useful not only technically but conceptually: you can position the proposed detector as bringing the maturity of ambulatory cardiac analysis into the EMU setting, where seizure timing from EEG provides an unusually strong external anchor. ²³

The third adjacent literature is **ECG-based seizure detection engineering**. This work is important because it shows what is now possible and what still fails. Jeppesen's modified HRV/Lorenz-plot work reported that the algorithm distinguished seizure tachycardia from exercise, that 13 of 17 patients had all seizures detected, and that all three secondarily generalized tonic-clonic seizures were detected more than half a minute before the tonic-clonic phase. Later adaptive approaches reduced false alarms materially, including a pediatric nocturnal detector that reported a 57% false-alarm reduction versus a nonadaptive one and patient-adaptive wearable approaches that lowered false alarms by roughly one-third. Very recent real-world wearable ECG benchmarks go a step further and show that ECG anomaly detection can achieve clinically meaningful sensitivity, but that the trade-off between sensitivity and false alarms remains the central deployment bottleneck. That combination of promise and limitation is highly relevant for your project: the signal contains useful seizure information, but patient adaptation, state adaptation, and quality gating are what make the difference between a publishable offline model and a clinically credible pipeline. ²⁴

What this means for your project

A literature-aligned architecture for the first project phase would be a **hierarchical pipeline** rather than a single monolithic classifier. The front end should do band-limited preprocessing, baseline-wander handling designed not to destroy ST content, R-peak detection, and multi-level signal-quality scoring. The second layer should generate patient- and state-aware baseline models from seizure-free EEG sessions and from the seizure-free cohort, stratified at minimum by wake versus sleep and ideally by EEG sleep stage. The third layer should compute features separately for rate, rhythm, morphology, and variability, with domain-specific quality masks. The final layer should apply change detection against matched baselines and output interpretable event flags rather than only a black-box score. That architecture is where the literature from epilepsy, PSG, ambulatory ECG, and signal-quality assessment converges most cleanly. ²⁵

The most defensible near-term claim for the project is not “automated SUDEP-risk diagnosis from single-lead ECG.” It is something more precise and much stronger scientifically: **automated, time-locked quantification of peri-ictal ECG change burden in concurrent EEG recordings**, beginning with rate and variability, extending to rhythm transitions, and then adding morphology where signal quality and lead geometry permit. If the first phase shows that seizure-free EMU recordings can support stable, sleep-aware personal baselines, then the second phase can credibly ask whether seizure windows deviate from those baselines in reproducible ways across domains. That framing is aligned with both the available evidence and the limitations of single-lead EMU ECG. ²⁶

If you want to be maximally well prepared for the meeting, the fastest reading sequence is this: start with the Zijlmans 2002 video-EEG paper for the foundational prevalence and timing data; then Chen 2014 for modifiers such as sleep, side, and seizure type; then Bruno 2018 for the pre-ictal meta-analysis; then the 2020 review on ECG changes beyond heart rate; then the 2021 review on ictal and interictal cardiac manifestations for the full clinical map; then the 2024 concurrent ECG-EEG review for why the field now wants standardization and automation; then the epilepsy HRV protocol paper for how to measure variability credibly; and finally the ambulatory ECG and VNS AutoStim literature for practical detection logic, thresholding, and false-alarm control. Read together, those papers will let you speak fluently about clinical importance, physiologic mechanisms, algorithm design, and the main reasons the field still does not yet have a dominant EMU ECG detector. ²⁷

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³ <https://www.sciencedirect.com/science/article/abs/pii/S1388245720300341>
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